Homework 4: Rainfall in 1970: likelihood kriging

Spatial statistics 2026–2027 · Professor: Dr. Lasinio

Mojtaba Alinejad

10 November 2026

Contents

1 My question and the 1970 data **1**

2 Exploratory analysis **2**
2.1 Locations and observed rainfall . . . 2
2.2 Trend and Gaussianity . . . 4

3 Model comparison by cross-validation **5**
3.1 Candidates and selection rule . . . 5
3.2 Fitting and prediction functions . . . 6
3.3 Rerunning the comparison for 1970 . . . 7

4 The fitted model and its diagnostics **8**
4.1 Variogram and directional checks . . . 9
4.2 Held-out residuals . . . 11

5 Prediction intervals: what they describe **11**

6 The 1970 prediction maps **12**

7 What I conclude for 1970 **14**

8 Sources and software **14**

1 My question and the 1970 data

Based on the feedback, I focus on **1970**, the year I analysed in Homework 2. I rerun the exploratory analysis, model comparison, diagnostics and grid predictions for this year. The aim is to predict annual rainfall and give an approximate 95% prediction interval at each supplied grid point.

Kriging combines a broad mean pattern with a correction based on nearby residuals. I use maximum likelihood to estimate each model and cross-validation to compare its predictions and intervals.

```
source_data <- new.env()
invisible(load("data/RainData.RData", envir = source_data))
dat <- subset(source_data$dati, anno == 1970)
grid <- source_data$grid
stopifnot(isTRUE(all.equal(grid,
  rbind(source_data$Basquota32, source_data$Calabriaquota32),
  check.attributes = FALSE)))

# I keep all three stations and correct only their regional labels.
region_fix <- c("4085" = "Basilicata", "4086" = "Basilicata",
```

```
                "4367" = "Calabria")
dat$Nome_Regione <- as.character(dat$Nome_Regione)
i <- dat$ID %in% as.numeric(names(region_fix))
dat$Nome_Regione[i] <- unname(region_fix[as.character(dat$ID[i])])
stopifnot(nrow(dat) == 189, all(dat$anno == 1970), !anyNA(dat),
  !anyNA(grid), all(dat$totanno > 0), !anyDuplicated(dat$ID),
  !anyDuplicated(dat[c("XUTM", "YUTM")]),
  !anyDuplicated(grid[c("XUTM", "YUTM")]))

add_coordinates <- function(d) {
  d$x <- d$XUTM/1000
  d$y <- d$YUTM/1000
  d$east <- (d$x-1100)/100
  d$north <- (d$y-4400)/100
  d$elev <- d$Quota/1000
  d
}
dat <- add_coordinates(dat)
grid <- add_coordinates(grid)
dat$rain <- dat$totanno
knitr::kable(data.frame(Stations = nrow(dat), Grid_points = nrow(grid),
  Mean_mm = mean(dat$rain), SD_mm = sd(dat$rain),
  Min_mm = min(dat$rain), Max_mm = max(dat$rain)), digits = 1)
```

Stations	Grid_points	Mean_mm	SD_mm	Min_mm	Max_mm
189	1005	829.1	317.3	289.8	1992.2

I retain all **189 observations** and the supplied **1,005-point land grid**. The original data file is unchanged. The matching grid is inside `RainData.RData`; the separately supplied `AllGrids.RData` contains marine data and is not used here. Only 1970 enters the analysis below.

As in Homework 2, I interpret rainfall as **mm**, elevation as **m**, and the supplied projected coordinates as **m**. The file lacks complete unit and coordinate-system metadata, so these remain assumptions. Distances use the common supplied coordinates divided by 1,000, in km. I do not assign an unverified EPSG code. The fixed centring and scaling of the trend covariates also apply inside validation folds.

Elevation is available at both the stations and the grid. I use it as an environmental predictor, with coordinates as possible additional trend terms. Latitude and longitude repeat location information. Regional labels describe the stations, but no corresponding regional factor is supplied for the grid.

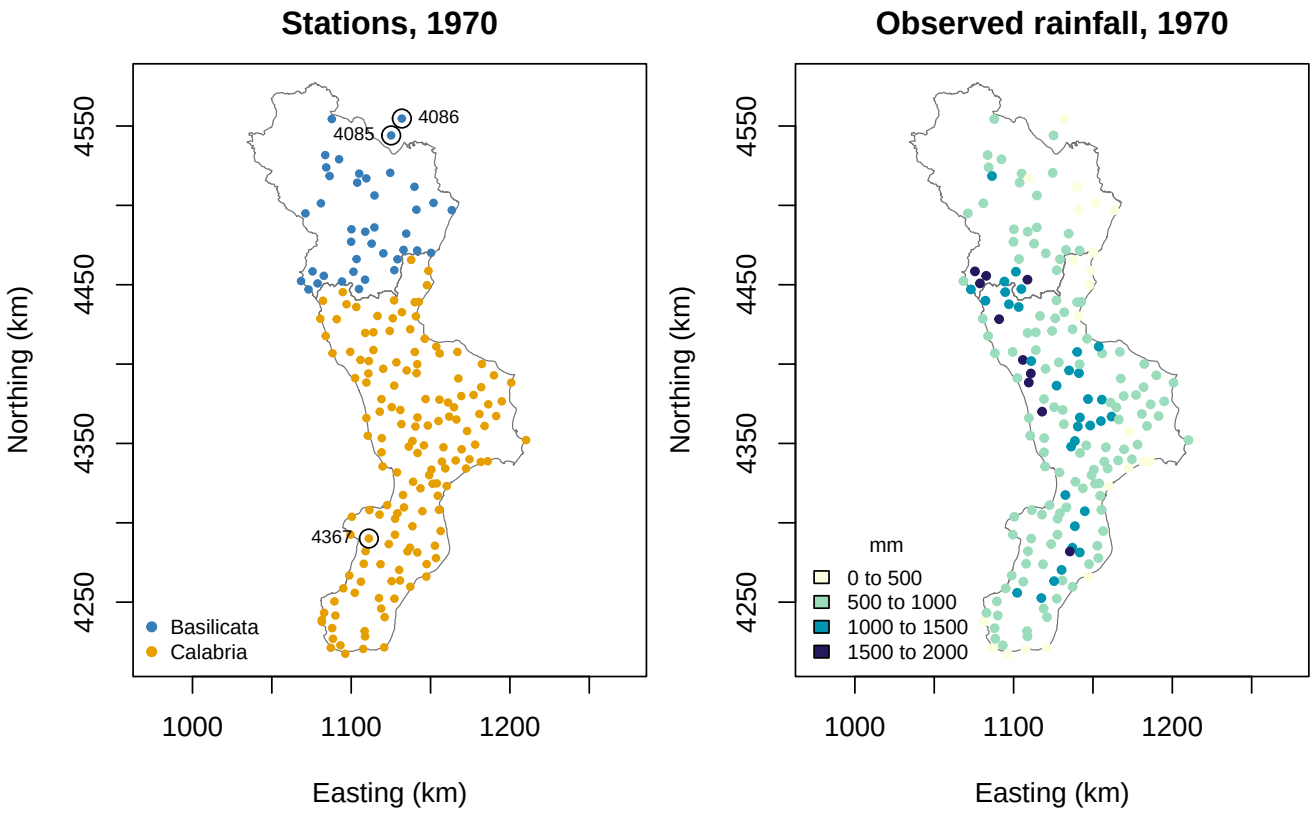
2 Exploratory analysis

2.1 Locations and observed rainfall

```
draw_outlines <- function() {
  lines(source_data$Bas$XUTM/1000, source_data$Bas$YUTM/1000,
        col = "grey45", lwd = 0.7)
  lines(source_data$Cal$XUTM/1000, source_data$Cal$YUTM/1000,
        col = "grey45", lwd = 0.7)
}
map_frame <- function(title) {
  plot(grid$x, grid$y, type = "n", asp = 1,
    xlim = range(c(grid$x, dat$x)), ylim = range(c(grid$y, dat$y)),
    xlab = "Easting (km)", ylab = "Northing (km)", main = title)
```

```
  draw_outlines()
}
colour_key <- function(breaks, colours) {
  legend("bottomleft",
    paste(head(breaks, -1), tail(breaks, -1), sep = " to "),
    fill = colours, title = "mm", bty = "n", cex = 0.75)
}
```

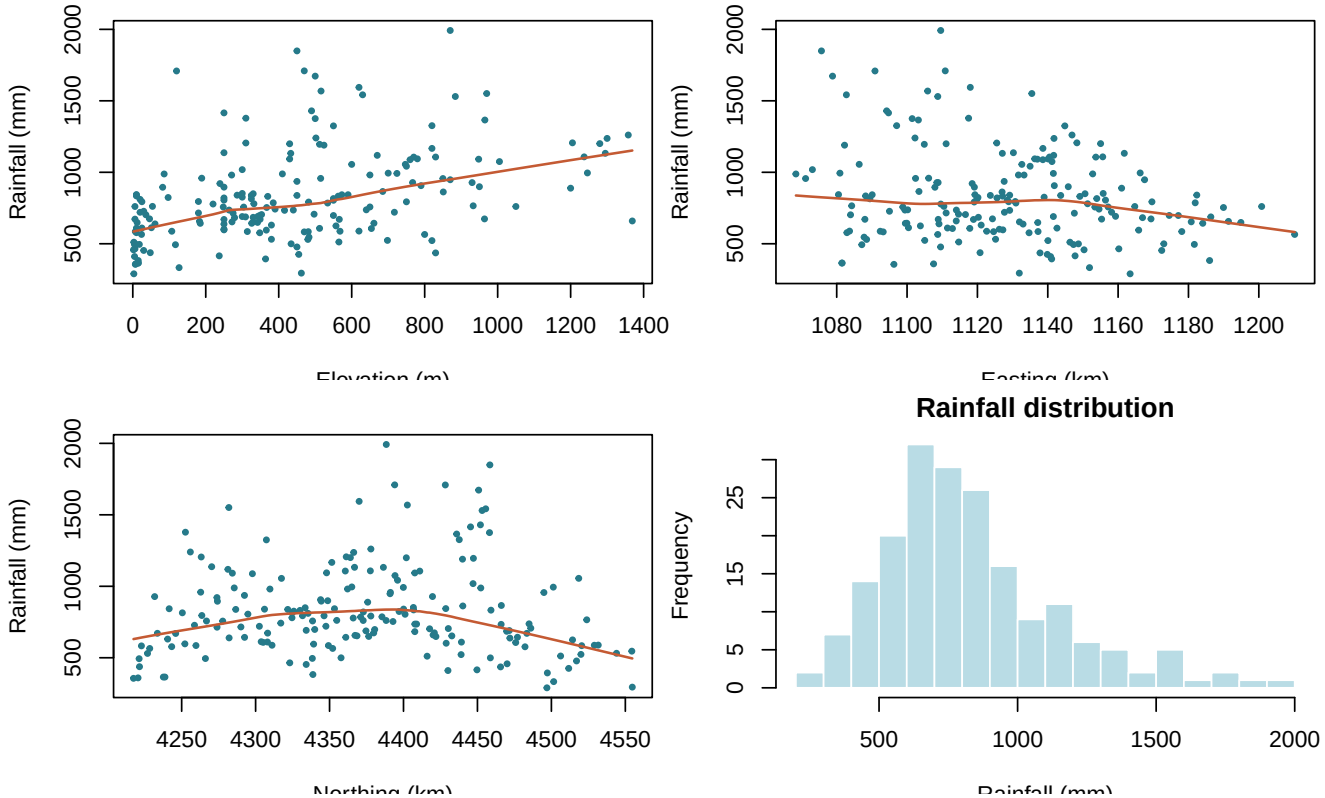
```
par(mfrow = c(1, 2), mar = c(4, 4, 2.5, 0.5))
map_frame("Stations, 1970")
region_col <- c(Basilicata = "#377EB8", Calabria = "#E69F00")
points(dat$x, dat$y, pch = 19, cex = 0.55,
       col = region_col[dat$Nome_Regione])
i <- match(as.numeric(names(region_fix)), dat$ID)
points(dat$x[i], dat$y[i], pch = 1, cex = 1.5)
text(dat$x[i], dat$y[i], dat$ID[i], pos = c(2, 4, 2), cex = 0.65)
legend("bottomleft", names(region_col), col = region_col,
       pch = 19, bty = "n", cex = 0.75)
map_frame("Observed rainfall, 1970")
breaks <- pretty(range(dat$rain), n = 5)
colours <- hcl.colors(length(breaks)-1, "YlGnBu", rev = TRUE)
points(dat$x, dat$y, pch = 19, cex = 0.65,
       col = colours[cut(dat$rain, breaks, include.lowest = TRUE)])
colour_key(breaks, colours)
```



The sampling region is elongated and the stations are unevenly spaced. There are wetter and drier groups of stations, so a constant mean is not an obvious description of the whole region. A visible pattern alone does not separate a spatial trend from residual spatial dependence.

2.2 Trend and Gaussianity

```
par(mfrow = c(2, 2), mar = c(3.5, 4, 2, 0.5))
for (v in c("Quota", "x", "y")) {
  label <- c(Quota = "Elevation (m)", x = "Easting (km)",
             y = "Northing (km)")[v]
  plot(dat[[v]], dat$rain, pch = 19, cex = 0.5, col = "#267A8A",
       xlab = label, ylab = "Rainfall (mm)")
  lines(lowess(dat[[v]], dat$rain), col = "#C45A33", lwd = 1.5)
}
hist(dat$rain, breaks = "FD", col = "#B9DCE5", border = "white",
     main = "Rainfall distribution", xlab = "Rainfall (mm)")
```



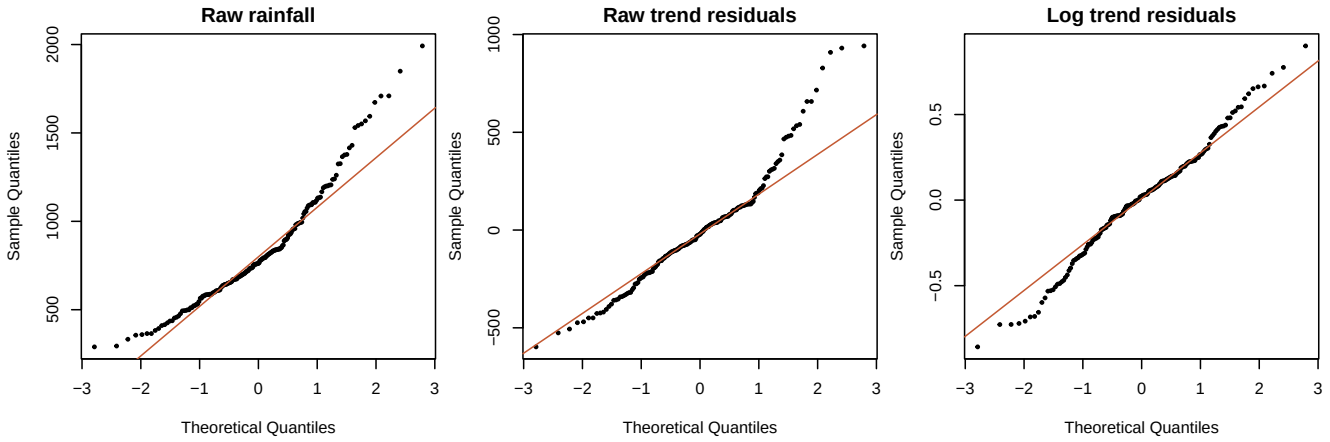
Rainfall is positively associated with elevation: the correlation is **0.43**. The coordinate plots also show broad geographical variation, especially lower rainfall towards the east. This supports comparing an elevation trend with an elevation-plus-coordinate trend. The association is descriptive; it is not a causal estimate of the effect of elevation.

```
raw_lm <- lm(rain ~ elev + east + north, data = dat)
log_lm <- lm(log(rain) ~ elev + east + north, data = dat)
skewness <- function(z) mean((z-mean(z))^3)/mean((z-mean(z))^2)^1.5
distribution_check <- data.frame(
  Quantity = c("Raw rainfall", "Raw trend residuals", "Log trend residuals"),
  Skewness = c(skewness(dat$rain), skewness(resid(raw_lm)),
               skewness(resid(log_lm))))
knitr::kable(distribution_check, digits = 2, row.names = FALSE)
```

Quantity	Skewness
Raw rainfall	1.05

Quantity	Skewness
Raw trend residuals	0.89
Log trend residuals	-0.10

```
par(mfrow = c(1, 3), mar = c(4, 4, 2, 0.5))
z <- list(dat$rain, resid(raw_lm), resid(log_lm))
for (j in 1:3) {
  qqnorm(z[[j]], main = distribution_check$Quantity[j],
         pch = 19, cex = 0.45)
  qqline(z[[j]], col = "#C45A33")
}
```



The raw rainfall and raw-scale trend residuals have a right tail. Removing the trend alone does not remove that asymmetry. The log-scale residuals are more balanced and closer to a normal Q–Q line. This supports considering a log transformation, although it does not prove a joint Gaussian model. I keep both response scales in the predictive comparison. Any selected log transformation will be used consistently in fitting, diagnostics and prediction.

3 Model comparison by cross-validation

3.1 Candidates and selection rule

For 1970 I model

$$T\{Y(s)\} = X(s)\beta + w(s) + \epsilon(s),$$

where T is either the identity or the natural logarithm. The trend $X(s)\beta$ describes the broad pattern; $w(s)$ is the remaining spatial process; and $\epsilon(s)$ has variance τ^2, the nugget. The nugget allows measurement error or small-scale variation. I assume stationary, isotropic residual covariance, and check that approximation later.

I compare two trends and three covariance families on each response scale: **12 candidates**. The exponential allows a rougher field; the spherical has a finite range; and Matérn with fixed $\kappa = 1.5$ allows a smoother field. Matérn $\kappa = 0.5$ equals the exponential, so it would be redundant.

```
trends <- list(Elevation = ~elev,
               Elevation_xy = ~elev + east + north)
candidates <- expand.grid(Scale = c("Raw", "Log"),
  Trend = names(trends),
  Covariance = c("exponential", "spherical", "matern"),
  stringsAsFactors = FALSE)
candidates$Model <- seq_len(nrow(candidates))
```

I use the same five folds for every candidate. Each model is fitted by **maximum likelihood using only training stations**, then predicts the held-out stations. No other year enters these fits. The fixed seed keeps the 1970 split reproducible. The analysis below is rerun without a cache.

Since uncertainty intervals are central to this homework, I select the lowest mean **95% Interval Score**, using RMSE to break an exact tie:

$$IS_{0.05}(l, u; y) = (u - l) + 40(l - y)1(y < l) + 40(y - u)1(y > u).$$

It penalises unnecessary width and observations outside the interval. It is a proper score for the central 95% predictive interval: its expected value is minimised by the true 2.5% and 97.5% predictive quantiles. Lower is better. Coverage alone can be inflated by making intervals too wide. I also report RMSE for the predictive mean, coverage, width and bias, all on the original rainfall scale. Scores are calculated at held-out stations; there are no observed grid values with which to score grid predictions.

These are **selection scores**, not an independent assessment of the whole selection procedure. Choosing the best of 12 models can make the winning score optimistic. Random folds mainly assess interpolation among the existing stations, not transfer to a large unsampled region.

3.2 Fitting and prediction functions

I use two initial range values and retain the converged fit with the higher likelihood. A failed start is counted. If neither start converges, the analysis stops. Convergence and multiple starts are useful checks, but cannot prove that a global maximum has been found.

```
design <- function(d, trend) {
  geoR::trend.spatial(trends[[trend]], d)
}

fit_one <- function(d, candidate) {
  z <- if (candidate$Scale == "Log") log(d$rain) else d$rain
  g <- geoR::as.geodata(data.frame(x = d$x, y = d$y, z = z,
    elev = d$elev, east = d$east, north = d$north), covar.col = 4:6)
  X <- design(d, candidate$Trend)
  s2 <- mean(lm.fit(X, z)$residuals^2)
  fits <- lapply(c(20, 80), function(start) {
    tryCatch(geoR::likfit(g, trend = trends[[candidate$Trend]],
      cov.model = candidate$Covariance, kappa = 1.5, fix.kappa = TRUE,
      ini.cov.pars = c(0.8*s2, start), nugget = 0.2*s2,
      fix.nugget = FALSE, lambda = 1, fix.lambda = TRUE,
      lik.method = "ML", limits = geoR::pars.limits(phi = c(0.5, 1500)),
      messages = FALSE, control = list(maxit = 2000)),
      error = function(e) NULL)
  })
  ok <- vapply(fits, function(m) {
    !is.null(m) && is.finite(m$loglik) &&
      m$info.minimisation.function$convergence == 0
  }, logical(1))
  if (!any(ok)) stop("No converged fit for this candidate.")
  valid <- fits[ok]
  fit <- valid[[which.max(vapply(valid, function(m) m$loglik, numeric(1)))]]
  list(fit = fit, g = g, X = X, rejected = sum(!ok))
}

predict_one <- function(f, locations, candidate) {
  p <- geoR::krige.conv(f$g,
    locations = as.matrix(locations[c("x", "y")]),
    krige = geoR::krige.control(type.krige = "OK", obj.model = f$fit,
      trend.d = f$X, trend.l = design(locations, candidate$Trend),
```

```
      micro.scale = 0),
    output = geoR::output.control(signal = FALSE, messages = FALSE))
  mu <- p$predict
  v <- p$krige.var
  stopifnot(all(is.finite(mu)), all(is.finite(v)), all(v > 0))
  lower <- mu - qnorm(0.975)*sqrt(v)
  upper <- mu + qnorm(0.975)*sqrt(v)
  if (candidate$Scale == "Log") {
    answer <- data.frame(Mean = exp(mu + v/2),
                         Lower = exp(lower), Upper = exp(upper))
  } else {
    answer <- data.frame(Mean = mu, Lower = lower, Upper = upper)
  }
  answer$mu <- mu
  answer$variance <- v
  stopifnot(all(is.finite(as.matrix(answer))))
  answer
}

score_predictions <- function(y, p) {
  interval_score <- (p$Upper-p$Lower) +
    40*pmax(p$Lower-y, 0) + 40*pmax(y-p$Upper, 0)
  data.frame(RMSE = sqrt(mean((p$Mean-y)^2)),
    IS95 = mean(interval_score),
    Coverage = 100*mean(y >= p$Lower & y <= p$Upper),
    Width = mean(p$Upper-p$Lower), Bias = mean(p$Mean-y))
}
```

The external trend matrices make this **universal kriging**, even though the `geoR` control option is named `OK`. I set `signal = FALSE` because the target is annual rainfall at a new station-like location, including the new observation's nugget variance.

3.3 Rerunning the comparison for 1970

```
set.seed(2026 + 1970)
fold <- sample(rep(1:5, length.out = nrow(dat)))
predictions <- scores <- vector("list", nrow(candidates))
rejected <- integer(nrow(candidates))
for (j in seq_len(nrow(candidates))) {
  p <- data.frame(Mean = rep(NA_real_, nrow(dat)), Lower = NA_real_,
                  Upper = NA_real_, mu = NA_real_, variance = NA_real_)
  for (k in 1:5) {
    test <- fold == k
    f <- fit_one(dat[!test, ], candidates[j, ])
    p[test, ] <- predict_one(f, dat[test, ], candidates[j, ])
    rejected[j] <- rejected[j] + f$rejected
  }
  predictions[[j]] <- p
  scores[[j]] <- score_predictions(dat$rain, p)
}
scores <- cbind(candidates, do.call(rbind, scores))
ranking <- order(scores$IS95, scores$RMSE)
best <- ranking[1]
runner_up <- ranking[2]
choice <- candidates[best, ]
knitr::kable(scores[ranking,
```

```
  c("Model", "Scale", "Trend", "Covariance", "IS95", "RMSE")],
  digits = 2, row.names = FALSE)
```

Model	Scale	Trend	Covariance	IS95	RMSE
6	Log	Elevation	spherical	1009.55	195.42
10	Log	Elevation	matern	1015.56	195.84
12	Log	Elevation_xy	matern	1016.90	196.33
4	Log	Elevation_xy	exponential	1020.25	196.16
2	Log	Elevation	exponential	1022.67	195.59
8	Log	Elevation_xy	spherical	1035.17	198.16
7	Raw	Elevation_xy	spherical	1312.73	204.40
1	Raw	Elevation	exponential	1315.89	201.02
3	Raw	Elevation_xy	exponential	1316.60	201.87
9	Raw	Elevation	matern	1323.15	201.38
11	Raw	Elevation_xy	matern	1326.96	202.15
5	Raw	Elevation	spherical	1329.29	204.88

For 1970 I select **Log rainfall, an elevation trend and spherical covariance**. Its mean Interval Score is **1009.55 mm** and RMSE is **195.42 mm**. The second-best interval score is **1015.56 mm**, for model **10**, the log-elevation Matérn model. The gap is only **6.01 mm**, or about **0.60%**. I therefore see the spherical model as a reasonable working choice, not strong evidence that its shape is better than every alternative.

The response scale matters more here: log-model Interval Scores range from **1009.6** to **1035.2 mm**, whereas raw-model scores range from **1312.7** to **1329.3 mm**. This agrees with the better-balanced log residuals in the exploratory plots. Adding east and north coordinates does not improve the best score once elevation and residual spatial covariance are included. A visible coordinate pattern therefore does not automatically justify extra trend terms.

```
chosen_scores <- scores[best, c("IS95", "RMSE", "Coverage", "Width", "Bias")]
knitr::kable(chosen_scores, digits = 2, row.names = FALSE)
```

IS95	RMSE	Coverage	Width	Bias
1009.55	195.42	92.59	663.77	4.75

```
cv <- predictions[[best]]
misses <- sum(dat$rain < cv$Lower | dat$rain > cv$Upper)
```

The intervals cover **92.59%** of held-out observations, with **14 of 189 observations outside**. This is below the nominal 95%, so the intervals are not perfectly calibrated. Their mean width is **663.77 mm**. The Interval Score is larger than this width because it also penalises misses. The mean prediction error is **4.75 mm**; positive values mean overprediction. These are results on one partition, with overlapping training samples, rather than a formal independent coverage test.

4 The fitted model and its diagnostics

```
final <- fit_one(dat, choice)
m <- final$fit
parameter_table <- data.frame(
  Parameter = c("Partial sill (log units squared)",
    "Nugget (log units squared)", "Range phi (km)", "Nugget share"),
  Estimate = c(m$cov.pars[1], m$nugget, m$cov.pars[2],
               m$nugget/(m$nugget+m$cov.pars[1])))
knitr::kable(parameter_table, digits = 4, row.names = FALSE)
```

Parameter	Estimate
Partial sill (log units squared)	0.1363
Nugget (log units squared)	0.0194
Range phi (km)	108.8744
Nugget share	0.1244

```
knitr::kable(data.frame(Term = colnames(model.matrix(trends[[choice$Trend]], dat)),
                        Coefficient = as.numeric(m$beta)),
             digits = 4, row.names = FALSE)
```

Term	Coefficient
(Intercept)	6.2701
elev	0.3968

```
range_bound <- m$cov.pars[2] < 0.51 | m$cov.pars[2] > 1499
stopifnot(m$info.minimisation.function$convergence == 0)
```

The selected spherical model has a finite residual-correlation range of about **108.9 km**. Its nugget is about **12.4%** of the total residual variance. This represents unstructured variation; it is not automatically evidence of poor measurements. The variance parameters refer to **log rainfall**, while the range still uses distances in km.

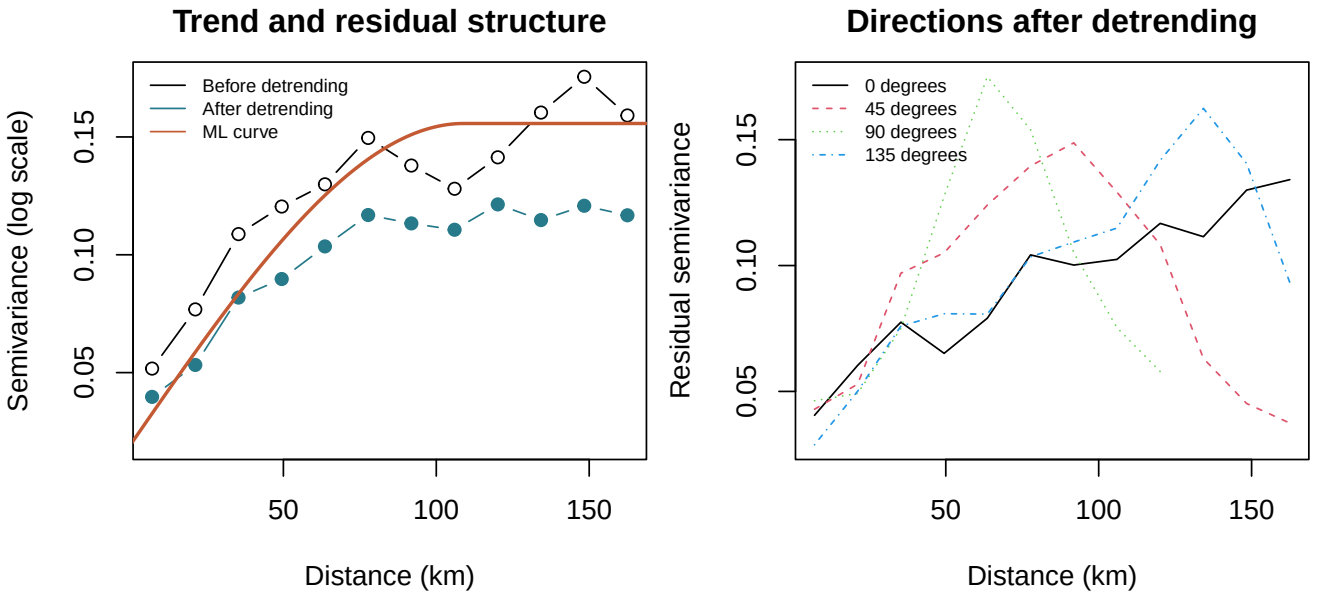
The fitted elevation coefficient corresponds to about **4.05%** higher median rainfall per 100 m of elevation in the trend, holding the spatial residual component fixed. It is an association within the model, not a causal effect. The broader coordinate trend did not win the predictive comparison.

There were **0 rejected optimisation starts** across validation and the final fit. The selected range is **away from the search boundaries**. I choose the trend by prediction quality, not simply by obtaining the smallest possible nugget.

4.1 Variogram and directional checks

```
cutoff <- max(dist(final$g$coords))/2
breaks <- seq(0, cutoff, length.out = 13)
v_raw <- geoR::variog(final$g, breaks = breaks, trend = "cte",
                      messages = FALSE)
v_res <- geoR::variog(final$g, breaks = breaks,
  trend = trends[[choice$Trend]], messages = FALSE)
directions <- lapply(c(0, 45, 90, 135), function(a) {
  geoR::variog(final$g, breaks = breaks, trend = trends[[choice$Trend]],
    direction = a*pi/180, tolerance = pi/8, messages = FALSE)
})
h <- seq(0, cutoff, length.out = 200)
gamma <- m$nugget + m$cov.pars[1] -
  geoR::cov.spatial(h, cov.model = m$cov.model,
                    cov.pars = m$cov.pars, kappa = m$kappa)
par(mfrow = c(1, 2), mar = c(4, 4, 2.5, 0.5))
plot(v_raw$u, v_raw$v, type = "b", pch = 1,
     ylim = range(c(v_raw$v, v_res$v, gamma)),
     xlab = "Distance (km)", ylab = "Semivariance (log scale)",
     main = "Trend and residual structure")
lines(v_res$u, v_res$v, type = "b", pch = 19, col = "#267A8A")
lines(h, gamma, col = "#C45A33", lwd = 2)
legend("topleft", c("Before detrending", "After detrending", "ML curve"),
       col = c("black", "#267A8A", "#C45A33"), lty = 1,
```

```
       cex = 0.7, bty = "n")
plot(v_res$u, v_res$v, type = "n",
     ylim = range(unlist(lapply(directions, function(v) v$v))),
     xlab = "Distance (km)", ylab = "Residual semivariance",
     main = "Directions after detrending")
for (j in 1:4) lines(directions[[j]]$u, directions[[j]]$v,
                     col = j, lty = j)
legend("topleft", c("0 degrees", "45 degrees", "90 degrees", "135 degrees"),
       col = 1:4, lty = 1:4, cex = 0.7, bty = "n")
```




```
knitr::kable(data.frame(Lag_km = v_res$u, Pairs = v_res$n,
                        Residual_semivariance = v_res$v),
             digits = 3, row.names = FALSE)
```

Lag_km	Pairs	Residual_semivariance
7.065	394	0.040
21.196	1019	0.053
35.327	1491	0.082
49.458	1648	0.090
63.589	1611	0.104
77.720	1529	0.117
91.850	1388	0.113
105.981	1273	0.111
120.112	1229	0.121
134.243	1087	0.115
148.374	980	0.121
162.505	874	0.117

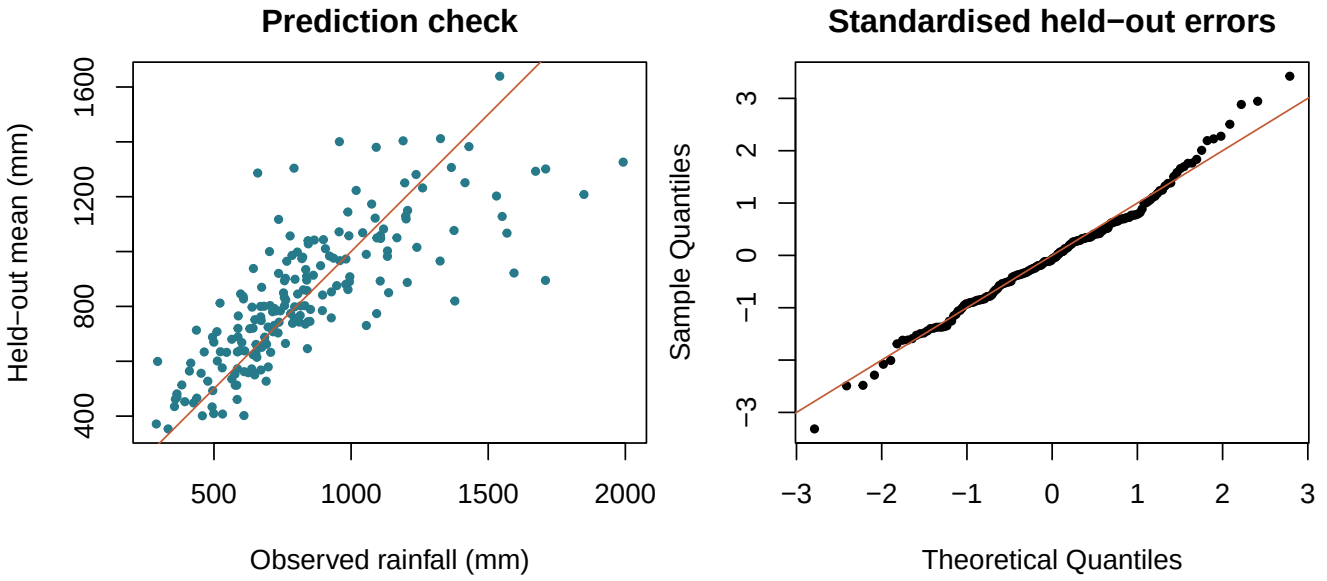
Both empirical curves are on the **log scale**. I use 12 equal-width bins up to **169.6 km**, half the largest station separation. Removing the elevation trend lowers the longer-distance semivariance, but the residual curve still rises at shorter distances. This supports modelling residual spatial structure after allowing for the mean pattern.

The ML spherical curve is a broad approximation; it does not pass through every empirical point. The fitted curve comes from the likelihood, not from least-squares fitting of these plotted points. For nonzero distance, $\gamma(h) = \tau^2 + \sigma^2\{1 - \rho(h)\}$; its value at the origin has a nugget discontinuity.

The directional curves differ, particularly at longer distances. Some directional bins have only **7 pairs**, so those estimates are noisy. The elongated sampling region also affects which directions and distances are represented. I treat stationarity and isotropy as working assumptions, not conclusions proved by these plots. An anisotropic model would be a possible extension, outside this candidate set.

4.2 Held-out residuals

```
observed_scale <- if (choice$Scale == "Log") log(dat$rain) else dat$rain
standardised <- (observed_scale-cv$mu)/sqrt(cv$variance)
par(mfrow = c(1, 2), mar = c(4, 4, 2.5, 0.5))
plot(dat$rain, cv$Mean, pch = 19, cex = 0.6, col = "#267A8A",
     xlab = "Observed rainfall (mm)", ylab = "Held-out mean (mm)",
     main = "Prediction check", asp = 1)
abline(0, 1, col = "#C45A33")
qqnorm(standardised, pch = 19, cex = 0.6,
       main = "Standardised held-out errors")
abline(0, 1, col = "#C45A33")
```



The predictions follow the broad rainfall pattern, with noticeable errors at individual stations. Most standardised errors lie fairly close to the normal reference line, but the tails are imperfect. Together with coverage below 95%, this is a reason to call the intervals approximate. The held-out errors share training observations, so I do not treat them as independent data for a formal normality test.

5 Prediction intervals: what they describe

The target is annual rainfall at an **unobserved location**, including small-scale or measurement variation. These are **approximate pointwise 95% prediction intervals from a maximum-likelihood model**. They include residual spatial variation, the nugget, and the universal-kriging correction for estimating the trend coefficients. They treat the estimated covariance parameters and selected model as fixed, so they omit that additional uncertainty. They are not a simultaneous 95% band over all grid points.

In a Bayesian analysis, a **credible interval** describes posterior uncertainty about a parameter, such as a regression coefficient, nugget, partial sill or range. A **posterior predictive interval** describes an unobserved value at a new location: it integrates parameter uncertainty and remaining spatial variation, and includes the nugget when predicting an observation rather than the latent field. Bayesian prediction maps should therefore label such intervals

posterior predictive intervals. Here I have fitted no posterior distribution, so I use *prediction interval* without the word *posterior*.

For the selected log model, if m and v are the kriging mean and prediction-error variance on the log scale, I use

$$\widehat{E(Y \mid \text{data})} = \exp(m + v/2), \qquad [l, u] = [\exp(m - 1.96\sqrt{v}),\ \exp(m + 1.96\sqrt{v})].$$

The factor $v/2$ matters: $\exp(m)$ alone is a predictive median, not the mean used for RMSE. I log rainfall explicitly and use `lambda = 1` in `geoR` to avoid transforming twice. The same back-transformation is used for held-out predictions and for the grid.

6 The 1970 prediction maps

```
mapped <- predict_one(final, grid, choice)
hull <- chull(dat$x, dat$y)
mapped$Outside_hull <- sp::point.in.polygon(
  grid$x, grid$y, dat$x[hull], dat$y[hull]) == 0
mapped$Outside_elevation <- grid$Quota < min(dat$Quota) |
                            grid$Quota > max(dat$Quota)
map_summary <- data.frame(Min_mean = min(mapped$Mean),
  Max_mean = max(mapped$Mean), Median_width = median(mapped$Upper-mapped$Lower),
  Outside_hull = sum(mapped$Outside_hull),
  Outside_elevation = sum(mapped$Outside_elevation))
knitr::kable(map_summary, digits = 1, row.names = FALSE)
```

Min_mean	Max_mean	Median_width	Outside_hull	Outside_elevation
316.3	2020.8	615	117	57

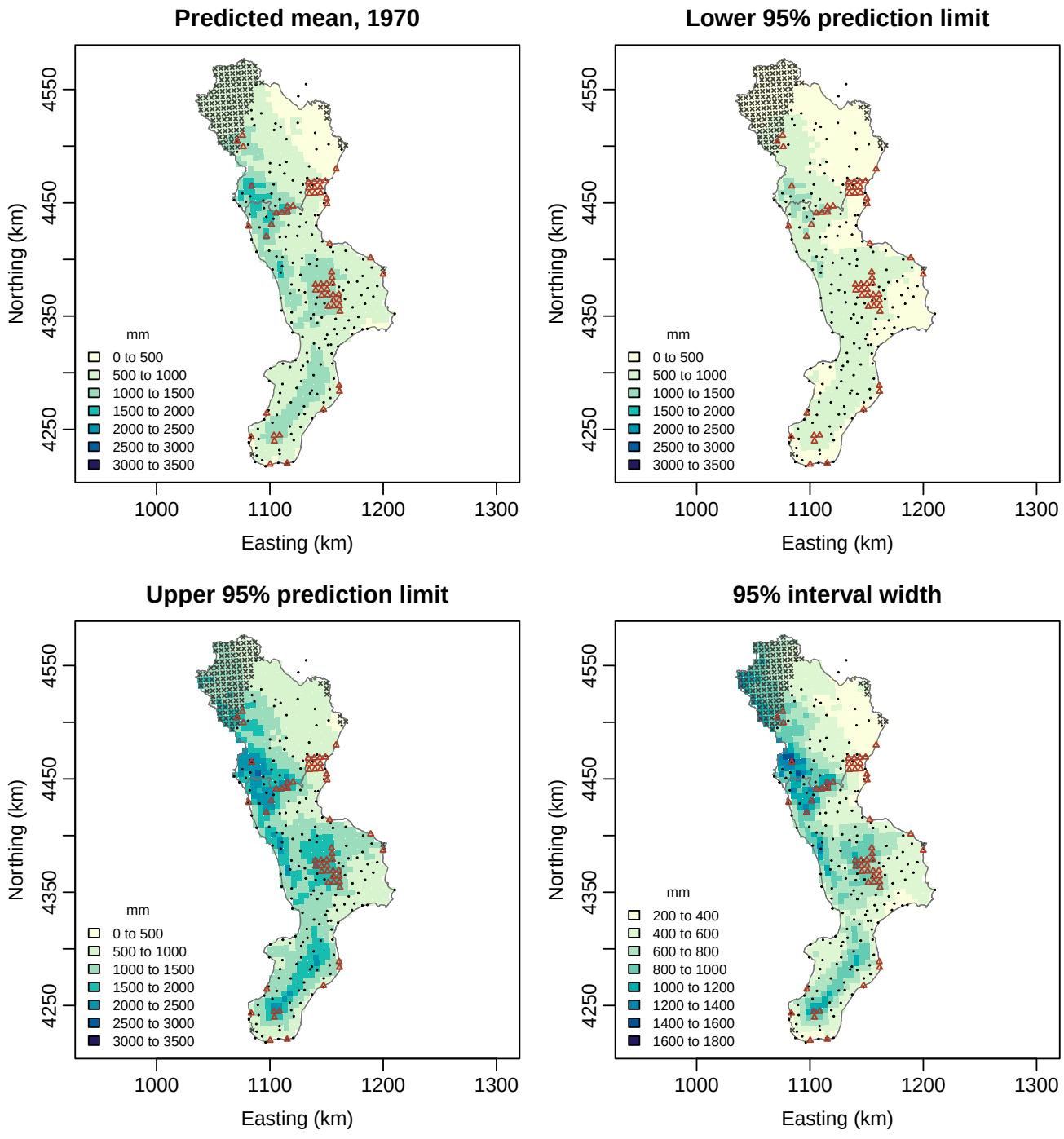
Predicted means range from **316.3 to 2020.8 mm**. The median interval width is **615.0 mm**. Higher predictions often follow elevated interior areas, with local changes from the spatial residual correction. I read these predictions together with their intervals.

There are **117 points outside the station convex hull** and **57 outside the observed elevation range**; the groups can overlap. Station elevations range from **2 to 1369 m**, while the grid reaches **1995 m**. High mountain predictions therefore rely partly on extrapolating the trend. A point inside the geographical region can still have an elevation outside the observed range.

```
draw_values <- function(values, breaks, title) {
  colours <- hcl.colors(length(breaks)-1, "YlGnBu", rev = TRUE)
  map_frame(title)
  points(grid$x, grid$y, pch = 15, cex = 0.55,
    col = colours[cut(values, breaks, include.lowest = TRUE)])
  draw_outlines()
  points(dat$x, dat$y, pch = 20, cex = 0.18)
  points(grid$x[mapped$Outside_hull], grid$y[mapped$Outside_hull],
         pch = 4, cex = 0.38, col = "grey25")
  points(grid$x[mapped$Outside_elevation], grid$y[mapped$Outside_elevation],
         pch = 2, cex = 0.48, col = "#B33A26")
  colour_key(breaks, colours)
}
```

```
breaks <- pretty(range(mapped[c("Mean", "Lower", "Upper")]), n = 6)
width_breaks <- pretty(range(mapped$Upper-mapped$Lower), n = 6)
par(mfrow = c(2, 2), mar = c(3.5, 3.5, 2.3, 0.5), mgp = c(2, 0.6, 0))
draw_values(mapped$Mean, breaks, "Predicted mean, 1970")
draw_values(mapped$Lower, breaks, "Lower 95% prediction limit")
```

```
draw_values(mapped$Upper, breaks, "Upper 95% prediction limit")
draw_values(mapped$Upper-mapped$Lower, width_breaks, "95% interval width")
```



The mean and two limits share a colour scale; width has its own scale. Black dots mark stations, crosses mark points outside their convex hull, and red triangles mark elevation extrapolation. All supplied grid points remain in the map. The log model keeps the prediction limits positive.

Wider intervals occur in parts of the northern edge and wetter interior. On the log scale, the same standard error produces a larger width in mm at a wetter location, so width is not simply a measure of station density. The mapped intervals also cannot capture uncertainty from an incorrect trend or covariance model, especially in extrapolation areas.

7 What I conclude for 1970

The predictive comparison supports log rainfall, an elevation trend and spherical residual covariance within this candidate set. Elevation accounts for part of the broad pattern, while kriging uses the remaining spatial association. The empirical and directional variograms still show features that a simple isotropic model cannot capture exactly.

The selected model has mean **95% Interval Score 1009.55 mm**, **RMSE 195.42 mm**, and **coverage 92.59%**. The coverage and tail diagnostics keep me cautious about interpreting the nominal 95% level. Repeating validation splits or using spatially separated validation would test stability more fully. Accounting for covariance-parameter uncertainty would also be a useful next step. Those extensions are not performed here.

8 Sources and software

I used the supplied lecture notes *Chapter1-2* for exploratory analysis and variograms, *Chapter3-5* for likelihood kriging and its prediction target, and *Chapter6-11*, Section 10.3, for the Interval Score. I use that score for central predictive intervals here, while retaining a likelihood framework. The `geoR` documentation supplies the function definitions and covariance parameterisations.

```
packages <- c("geoR", "sp", "knitr", "rmarkdown")
knitr::kable(data.frame(Component = c("R", packages),
  Version = c(as.character(getRversion()),
    vapply(packages, function(p) as.character(packageVersion(p)),
           character(1)))), row.names = FALSE)
```

Component	Version
R	4.5.0
geoR	1.9.6
sp	2.2.0
knitr	1.50
rmarkdown	2.29